

Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and sales channels (online vs. offline). They are particularly interested in uncovering which factors, such as discounts, reviews, seasons, or payment preferences, drive consumer decisions and repeat purchases.

You are tasked with analyzing the company's consumer behavior dataset to answer the following overarching business question:

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

Deliverables

1. Data Preparation & Modeling (Python): Clean and transform the raw dataset for analysis.
2. Data Analysis (SQL): Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.
3. Visualization & Insights (Power BI): Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
4. Report and Presentation: Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.

Project Report: Customer Shopping Behavior Analysis

Executive Summary

This project analyzes the shopping behavior of 3,900 customers to extract actionable business insights regarding purchasing trends, demographics, and product performance. An end-to-end data pipeline was utilized: **Python** was used for data cleaning and feature engineering, **SQL** was utilized to query specific business metrics, and **Power BI** was used to build an interactive dashboard.

Key findings reveal that **Young Adults** are the highest revenue-driving demographic, **Clothing** is the dominant product category, and there is a significant opportunity to increase recurring revenue, as **73% of customers are not currently subscribed** to the business's services.

KPIs required for SQL Queries

- Total revenue generated by male vs. female customers (grouped by gender).
- Customers who used a discount but still spent more than the average purchase amount.
- Top 5 products with the highest average review rating.
- Average Purchase Amount comparison between Standard and Express Shipping.
- Do subscribed customers spend more? — Compares average spend and total revenue between subscribers and non-subscribers.
- Top 5 products with the highest percentage of purchases where a discount was applied.
- Customer segmentation into New, Returning, and Loyal based on total number of previous purchases + count of customers in each segment.
- Top 3 most purchased products within each category (using window functions).
- Are repeat buyers (>5 previous purchases) also likely to subscribe? — Counts repeat buyers by subscription status.
- Revenue contribution of each age group.

Chart Requirement

Slicers / Filters

1. **Gender:** Female, Male
2. **Category:** Accessories, Clothing, Footwear, Outerwear
3. **Shipping Type:** 2-Day Shipping, Express, Free Shipping, Next Day Air, Standard, Store Pickup
4. **Subscription Status:** Yes, No

Charts & Visualizations

1. **% of Customers by Subscription Status (Donut Chart) :-** Shows the proportion of customers who have a subscription (Yes) versus those who don't (No). Current split: Yes = 27%, No = 73%. This helps measure subscription adoption and loyalty potential.
2. **Revenue by Category (Vertical Bar Chart) :-** Displays total revenue generated by each product category (Clothing highest, followed by Accessories, Footwear, and Outerwear). It highlights which categories contribute the most to overall revenue.
3. **Sales by Category (Vertical Bar Chart) :-** Shows the total number of sales/orders by product category. It reveals which categories are most popular in terms of purchase volume.
4. **Revenue by Age Group (Horizontal Bar Chart) :-** Compares total revenue across age groups (Young Adult highest, followed by Middle-aged, Adult, and Senior). This identifies the most valuable age segments by spending power.

5. **Sales by Age Group (Horizontal Bar Chart) :-** Displays the total number of purchases/orders by age group. It helps understand purchasing frequency and activity levels across different age demographics.

1. Introduction

- **Objective:** To understand customer purchasing patterns, segment the customer base, and identify key drivers of revenue to inform future marketing and sales strategies.
- **Dataset:** The raw dataset consists of 3,900 records and 18 columns, detailing customer demographics (Age, Gender, Location), purchase details (Category, Item, Amount), and transaction behaviors (Frequency, Discount Usage, Subscription Status).
- **Tech Stack Used:**
 - **Python (Pandas):** Data manipulation, handling missing values, and feature engineering.
 - **SQL:** Relational database querying for targeted business insights.
 - **Power BI:** Data visualization and interactive dashboard creation.

2. Phase 1: Data Cleaning & Manipulation (Python)

The raw CSV file required preparation before analysis. Using the Pandas library in Python, the following transformations were performed:

1. **Handling Missing Data:** Identified missing values in the Review Rating column (37 null values). These were imputed dynamically based on the median/mean rating of their respective product Category.
2. **Standardizing Column Names:** Converted all column names to lowercase and replaced spaces with underscores (e.g., Customer ID to customer_id) for seamless SQL querying.
3. **Feature Engineering:**
 - Created an **age_group** column using `pd.qcut` to categorize customers into four logical segments: Young Adult, Adult, Middle-aged, and Senior.
 - Created a **purchase_frequency_days** column by mapping categorical text (e.g., "Weekly", "Fortnightly") into numerical day values (7, 14, etc.) to allow for mathematical analysis of purchase intervals.
4. **Removing Redundancy:** Discovered that the discount_applied and promo_code_used columns were identical (100% match). To optimize the dataset, the promo_code_used column was dropped.

3. Phase 2: Exploratory Data Analysis & Insights (SQL)

The cleaned dataset was imported into a SQL database to answer specific business questions. Complex queries utilizing aggregation, filtering, and window functions were written to extract the following insights:

- **Customer Segmentation:** Customers were segmented based on their previous purchase history into 'New' (1 purchase), 'Returning' (2-10 purchases), and 'Loyal' (>10 purchases) to understand retention rates.
- **Subscription vs. Non-Subscription Spend:** Analyzed the average spend and total revenue comparing subscribed customers versus non-subscribed customers to determine the financial value of the subscription program.
- **Product Performance:** Identified the Top 5 products with the highest average review ratings, as well as the Top 5 products most frequently bought using a discount.
- **Category Leaders:** Utilized the ROW_NUMBER() OVER (PARTITION BY ...) window function to identify the top 3 most purchased products within *each* specific category.
- **Shipping & Spend:** Compared average purchase amounts between "Standard" and "Express" shipping types to see if customers who spend more also opt for faster shipping.

4. Phase 3: Data Visualization (Power BI)

To make the data accessible to stakeholders, an interactive dashboard was built in Power BI.

Key Performance Indicators (KPIs):

- **Total Customers:** 3.9K
- **Average Purchase Amount:** \$59.76
- **Average Review Rating:** 3.75 / 5.0

Dashboard Visual Insights:

- **Subscription Breakdown:** A donut chart reveals a major business gap: **73% of customers do not have a subscription**, while only 27% do.
- **Revenue & Sales by Category:** Bar charts indicate that **Clothing** is by far the highest-performing category in both total sales volume (~1,700 items) and total revenue (>100K). It is followed by Accessories and Footwear, with Outerwear performing the lowest.
- **Demographic Performance:** **Young Adults** generate the highest revenue and transaction volume. While Seniors represent a healthy portion of the sales count, Middle-aged and Young Adult shoppers generate higher total revenue.

5. Key Findings & Business Recommendations

Based on the end-to-end analysis, the following actionable recommendations are proposed:

1. **Prioritize Subscription Conversion:** With 73% of the customer base unsubscribed, there is a massive opportunity for recurring revenue. *Recommendation:* Offer a one-time discount on the highest-performing category (Clothing) exclusively for customers who sign up for a new subscription.
2. **Targeted Marketing by Age:** "Young Adults" are the most valuable demographic. *Recommendation:* Allocate a higher percentage of the digital marketing and social media ad budget toward younger demographics, specifically highlighting trending "Clothing" and "Accessories."
3. **Boost Low-Performing Categories:** "Outerwear" is the lowest-performing category in both volume and revenue. *Recommendation:* Create bundled promotions (e.g., "Buy a piece of Outerwear, get 20% off Footwear") or analyze seasonal trends to push Outerwear heavier during the Fall/Winter months.

Leverage High-Rated Products: Based on the SQL analysis of the top-rated products, *Recommendation:* Feature these specific 5-star items on the front page of the website or in email marketing campaigns to drive guaranteed customer satisfaction.

SQL Queries

```
create database customer_behavior;
use customer_behavior;
select * from customer limit 20;
```

customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status	shipping_type	discount_applied	previous_purchases	payment_method
1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	14	Venmo
2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	2	Cash
3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	23	Credit Card
4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	49	PayPal
5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	31	PayPal
6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	14	Venmo
7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	49	Cash

KPI's

Q1. What is the total revenue generated by male vs. female customers?

```
select gender, sum(purchase_amount) as revenue from customer group by gender;
```

gender	revenue
Male	134364
Female	63629

Q2. Which customers used a discount but still spent more than the average purchase amount?

```
select customer_id, purchase_amount from customer where discount_applied = 'Yes' and purchase_amount >= (select avg(purchase_amount) from customer) limit 14;
```

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81

20	90
22	62
24	88
32	79
33	67
35	91

Q3. Which are the top 5 products with the highest average review rating?

```
select item_purchased, round(avg(review_rating),2) as "Average_Product_Rating" from customer group by item_purchased order by avg(review_rating) desc limit 7;
```

item_purchased	Average_Product_Rating
Sandals	3.86
Gloves	3.85
Boots	3.84
Hat	3.8
Skirt	3.8
Sweater	3.79
Jewelry	3.78

Q4. Compare the average Purchase Amounts between Standard and Express Shipping.

```
select shipping_type, round(avg(purchase_amount),2) from customer where shipping_type in ('Standard','Express') group by shipping_type;
```

shipping_type	round(avg(purchase_amount),2)
Express	61.00
Standard	58.53

Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

```
select subscription_status, count(customer_id) as total_customer, round(avg(purchase_amount),2) as Avg_Spend, round(sum(purchase_amount),2) as total_revenue from customer group by subscription_status order by total_revenue, avg_spend Desc;
```

subscription_status	total_customer	Avg_Spend	total_revenue
Yes	899	59.28	53289
No	2417	59.87	144704

Q6. Which 5 products have the highest percentage of purchases with discounts applied?

```
select item_purchased, round(sum(case when discount_applied='Yes' then 1 Else 0 End) /count(*) * 100,2) as discount_rate from customer group by item_purchased order by discount_rate desc limit 5;
```

item_purchased	discount_rate
Sneakers	52.03
Hoodie	49.22
Coat	48.95
Jewelry	48.63
Hat	48.51

Q7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

```
with customer_type as(
    select customer_id, previous_purchases,
    case
        when previous_purchases=1 then 'New'
        when previous_purchases between 2 and 10 then 'Returning'
        else 'Loyal' end as customer_segment
    from customer)

    select customer_segment, count(*) as "Number of Customers" from
customer_type group by customer_segment;
```

customer_segment	Number of Customers
Loyal	2660
Returning	591
New	65

Q8. What are the top 3 most purchased products within each category?

```
with item_counts AS (
    select
        category,
        item_purchased,
        COUNT(customer_id) AS total_orders,
        ROW_NUMBER() OVER (PARTITION BY category ORDER BY
COUNT(customer_id) DESC) AS item_rank
    from customer
    group by category, item_purchased
)
select
    item_rank, category, item_purchased, total_orders from item_counts
where item_rank <= 3;
```

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	146
2	Accessories	Sunglasses	144
3	Accessories	Scarf	136
1	Clothing	Shirt	149
2	Clothing	Pants	146
3	Clothing	Sweater	142
1	Footwear	Sandals	133
2	Footwear	Shoes	126

3	Footwear	Boots	125
1	Outerwear	Coat	143
2	Outerwear	Jacket	134

Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```
select subscription_status, count(customer_id) as repeat_buyers from
customer where previous_purchases >5 group by subscription_status;
```

subscription_status	repeat_buyers
Yes	815
No	2143

Q10. What is the revenue contribution of each age group?

```
select age_group, sum(purchase_amount) as total_revenue from customer
group by age_group order by total_revenue desc;
```

age_group	total_revenue
Young Adult	52189
Middle-aged	50406
Senior	47902
Adult	47496

Dashboard Image :-

